CS182: Introduction to Machine Learning
Lecture 03: Probability and Statistics

Lecture Quiz and Answer Copy

Scope: Probability_Statistics.pdf. Independent practice material based on the official 2026 Fall lecture.

Total: 100 marks. **Suggested time:** 120 minutes.

Name: _____ **Student ID:** _____

Section	Questions	Each	Marks
A. Misconceptions: True/False	12	1	12
B. Multiple response	12	2	24
C. Proofs and formula derivations	4	10	40
D. Applications	2	12	24
Total	30		100

Scoring. In Section B each item has exactly four options, of which one, two, or three are correct. The complete correct set earns 2 marks; a nonempty proper subset earns 1 mark; any incorrect option or no selection earns 0. T/F answers earn 1 mark only when correct. Show reasoning in Sections C and D; part marks are available for valid intermediate work. Equivalent correct methods receive credit.

Instructions. Answer in English. Use natural logarithms. Exact values and justified approximations are accepted. All vectors are columns unless stated otherwise. Use local assumptions, label conventions, and tie rules.

Answer sheet — Section A

1	2	3	4	5	6

7	8	9	10	11	12

Answer sheet — Section B

13	14	15	16	17	18

19	20	21	22	23	24

Section A. True/False 12 marks

1. If two events have positive probabilities and are mutually exclusive, then they are independent.

2. For iid nonnormal observations with finite positive variance, the central limit theorem makes the standardized sample mean exactly standard normal at every finite sample size.

3. If $T \sim t_3$, its variance is one because it is a standardized ratio.

4. If two variables each have a normal marginal distribution and have zero covariance, they must be independent.

5. The identity $E(X + Y) = E(X) + E(Y)$ requires independence.

6. The weak law of large numbers says that, for a sufficiently large finite sample, the sample mean equals the population mean with probability one.

7. For iid observations with finite variance σ^2, the variance of their sample mean is σ^2/n.

8. For a random variable with a probability density, the density value $p(x)$ may exceed one even though $P(X = x) = 0$.

9. For finite-variance variables, $\mathrm{Var}(X - Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) - 2\,\mathrm{Cov}(X, Y)$.

10. For iid $N(\mu, \sigma^2)$ observations with $n \geq 2$, $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$, where $S^2 = \sum_i (X_i - \bar{X})^2/(n-1)$.

11. If $Z_1, \ldots, Z_k$ are independent standard normal variables, then $\sum_i Z_i^2$ has a chi-square distribution with k degrees of freedom.

12. For events with $P(A), P(B) > 0$, the identity $P(A \mid B) = P(B \mid A)$ holds even when the prior probabilities differ.

Section B. Multiple Response 24 marks

Select all correct options. Each question has 1–3 correct options.

13. Let $X_1, \dots, X_n$ be iid with mean 7 and finite variance 9. Select every justified conclusion.

(A) For every finite n, $\sum_i X_i \sim N(7n, 9n)$.

(B) $P(|\bar{X}_n - 7| > 0.1) \to 0$ as $n \to \infty$.

(C) $\mathrm{Var}(\bar{X}_n) = 9/n^2$.

(D) For large n, $\bar{X}_n$ is approximately $N(7, 9/n)$.

14. Select every valid probability identity. Assume all conditional probabilities displayed are defined.

(A) $P(A \cap B \cap C) = P(A)P(B \mid A)P(C \mid B)$.

(B) $P(A \mid B) = P(A \cap B)/P(A)$.

(C) If A_1, A_2 partition the sample space, then $P(B) = P(B \mid A_1) + P(B \mid A_2)$.

(D) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

15. A joint density is $p(x, y) = 2$ on $0 < x < y < 1$, and zero elsewhere. Select every correct statement.

(A) $p_X(x) = 2(1 - x)$ for $0 < x < 1$.

(B) X and Y are independent because their joint density is constant on its support.

(C) The density integrates to one.

(D) For $0 < y < 1$, $p_{X|Y}(x \mid y) = 1$ on $0 < x < y$.

16. Let X take values $-1, 0, 1$ with equal probabilities and let $Y = X^2$. Select every correct claim.

(A) $E[X] = 0$ and $E[Y] = 2/3$.

(B) X and Y are independent.

(C) $\mathrm{Corr}(X, Y)$ is undefined because $E[X] = 0$.

(D) $\mathrm{Cov}(X, Y) = 2/9$.

Section B. Multiple Response 24 marks

Select all correct options. Each question has 1–3 correct options.

17. The counts (N_1, N_2, N_3) come from 10 independent categorical trials with probabilities $(0.2, 0.3, 0.5)$. Select every correct statement.

(A) Each count is Bernoulli distributed.

(B) $N_1 + N_2 + N_3 = 10$ with probability one.

(C) N_1, N_2, N_3 are independent because the ten trials are independent.

(D) $P(N_1 = 2, N_2 = 3, N_3 = 5) = \frac{10!}{2!3!5!}(0.2)^2(0.3)^3(0.5)^5$.

18. For a variable with finite second moment, select every correct moment formula.

(A) $\mathrm{Var}(3X - 7) = 9\,\mathrm{Var}(X)$.

(B) $E[X^2] = (EX)^2$ for every random variable.

(C) $\mathrm{Cov}(2X + 1, -Y + 4) = -2\,\mathrm{Cov}(X, Y)$.

(D) $E[(X - EX)^2] = E[X^2] - (EX)^2$.

19. A nonsingular bivariate Gaussian has mean $(1, -2)^T$ and covariance $\Sigma = \begin{pmatrix} 4 & 0 \\ 0 & 9 \end{pmatrix}$. Select every correct statement.

(A) The two coordinates are independent.

(B) The exponent contains $(x_1 - 1)^2/4 + (x_2 + 2)^2/9$, multiplied by $-1/2$.

(C) The density normalization contains $|\Sigma|^{1/2} = 6$.

(D) The density at its mean equals the probability that the vector equals its mean.

20. A count K records successes in 20 independent Bernoulli trials with success probability 0.15. Select every correct formula.

(A) The fraction $K/20$ has variance $2.55/20$.

(B) $P(K = 2) = \binom{20}{2}(0.15)^2(0.85)^{18}$.

(C) $E[K] = 3$.

(D) $\mathrm{Var}(K) = 3$.

Section B. Multiple Response 24 marks

Select all correct options. Each question has 1–3 correct options.

21. For iid $X_i \sim N(\mu, \sigma^2)$, $n \geq 3$, let $S^2 = \sum_i (X_i - \bar{X})^2/(n-1)$. Select every correct distribution statement.

(A) $\bar{X}$ and S^2 are independent.

(B) $\sqrt{n}(\bar{X} - \mu)/S \sim t_n$.

(C) $nS^2/\sigma^2 \sim \chi^2_n$.

(D) $\sqrt{n}(\bar{X} - \mu)/\sigma \sim N(0,1)$.

22. Let $T \sim t_5$ and $W = 4 + 2T$. Select every correct claim.

(A) T is exactly standard normal because its degrees of freedom exceed two.

(B) W has a general t distribution with location 4 and scale 2.

(C) $\mathrm{Var}(W) = 4$.

(D) $E[W] = 4$.

23. Two independent fair bits U, V are drawn. Let $A = \{U = 1\}$, $B = \{V = 1\}$, and $C = \{U = V\}$. Select every correct claim.

(A) $P(C \mid A \cap B) = 1$.

(B) Pairwise independence always implies independence of every triple intersection.

(C) A and C are independent.

(D) A, B, C are mutually independent.

24. A binary joint distribution is $p(0,0) = 0.30$, $p(0,1) = 0.10$, $p(1,0) = 0.20$, $p(1,1) = 0.40$. Select every correct conclusion.

(A) $P(X = 1) = 0.60$.

(B) $P(X = 0 \mid Y = 0) = 0.30$.

(C) X and Y are independent.

(D) $P(Y = 1 \mid X = 1) = 2/3$.

Section C. Proofs and Formula Derivations

25. Deriving total probability, Bayes' rule, and posterior odds **[10 marks]**

(a) Let $A_1, \ldots, A_k$ be a finite partition with $P(A_i) > 0$. Starting from disjoint events and the product rule, derive a formula for $P(B)$. State the additional condition needed to condition on B.**[3 marks]**

(b) Use the result to derive $P(A_i \mid B)$, and prove that these posterior probabilities sum to one. **[3 marks]**

(c) For two hypotheses H_1, H_0 with positive priors, derive posterior odds after observing events E and F. Assume the relevant likelihoods are positive. Show when the evidence contribution factors into $P(E \mid H_i)P(F \mid H_i)$. **[4 marks]**

Student response

(a)

(b)

(c)

Section C. Proofs and Formula Derivations

26. Variance formulas and the effect of shared noise **[10 marks]**

(a) For finite-second-moment variables, derive $\mathrm{Cov}(X, Y) = E[XY] - E[X]E[Y]$ and the variance of $aX + bY + c$. **[3 marks]**

(b) Let $X_1, \ldots, X_n$ have common mean μ and finite variance σ^2. Derive the mean and variance of $\bar{X}_n$ in terms of all pairwise covariances. Specialize to independent observations. **[4 marks]**

(c) Suppose measurements have the form $X_i = \mu + B + \epsilon_i$, where $B, \epsilon_1, \ldots, \epsilon_n$ are independent, centered, and have variances $\tau^2, \sigma_\epsilon^2, \ldots, \sigma_\epsilon^2$. Derive $\mathrm{Var}(\bar{X}_n)$ and explain why the iid variance formula would fail. **[3 marks]**

Student response

(a)

(b)

(c)

Section C. Proofs and Formula Derivations

27. Constructing the binomial distribution from trials **[10 marks]**

(a) Let $I_1, \ldots, I_n$ be independent Bernoulli variables with common success probability $p \in (0, 1)$. For $K = \sum_i I_i$, derive $P(K = k)$ by counting disjoint outcomes. **[3 marks]**

(b) Derive the mean and variance of one Bernoulli variable directly from its two possible values. Then derive $E[K]$ without differentiating a generating function. **[3 marks]**

(c) Derive $\mathrm{Var}(K)$ and $\mathrm{Var}(K/n)$. Identify precisely where independence is used, and state the variance formula if the indicators are dependent but retain the same marginal success probability.**[4 marks]**

Student response

(a)

(b)

(c)

Section C. Proofs and Formula Derivations

28. An exact Gaussian mean and residual decomposition **[10 marks]**

(a) Let X_i be iid $N(\mu, \sigma^2)$, $\sigma > 0$, $n \geq 2$, and set $Z_i = (X_i - \mu)/\sigma$. Choose an orthogonal matrix A whose first row is $(1, \ldots, 1)/\sqrt{n}$, and let $Y = AZ$. Using the joint standard normal density, prove that the coordinates of Y are independent standard normals. **[3 marks]**

(b) Express $\sqrt{n}(\bar{X} - \mu)/\sigma$ and $\sum_i (X_i - \bar{X})^2/\sigma^2$ using these coordinates. Derive the chi-square law and prove independence of the sample mean and sample variance. **[4 marks]**

(c) For $S^2 = \sum_i (X_i - \bar{X})^2/(n-1)$, derive the exact distribution of $\sqrt{n}(\bar{X} - \mu)/S$. Explain why the same exact assertion does not follow merely from iid sampling with finite variance. **[3 marks]**

Student response

(a)

(b)

(c)

Section D. Applications

29. Two inspections and the base rate of defective components **[12 marks]**

(a) A manufacturer draws components from a stable production stream. A component is defective with probability 0.03. One inspection flags a defective component with probability 0.90, and flags a nondefective component with probability 0.04. Compute the flag probability and the probability a flagged component is defective. **[4 marks]**

(b) The component is inspected twice. Assume the inspection outcomes are independent conditional on whether that component is defective, with the same sensitivity and false-flag probability each time. Compute the probability of two flags and the defect probability given two flags. Explain the factorization you used. **[4 marks]**

(c) A separate batch contains 200 independently drawn components, each inspected once, with independent inspection noise across components. Let K be its flag count. State its exact distribution, mean, variance, and the variance of its flag fraction $K/200$. **[4 marks]**

Student response

(a)

(b)

(c)

Section D. Applications

30. Exact inference for a normally distributed calibration error **[12 marks]**

(a) Nine independent calibration measurements, in millimeters, are

$$498,\ 502,\ 501,\ 499,\ 500,\ 503,\ 497,\ 500,\ 500.$$

Assume an iid $N(\mu, \sigma^2)$ model with both parameters unknown. Compute $\bar{x}$, the residual sum of squares, the unbiased sample variance s^2, and the estimated standard error of the mean.**[4 marks]**

(b) Compute the t statistic for the target mean $\mu_0 = 499$. Using $t_{8,0.975} = 2.306$, derive and calculate a central 95% confidence interval for μ. State the exact sampling assumption behind this interval.**[4 marks]**

(c) For the reference variance $\sigma_0^2 = 4$, compute the chi-square statistic. Use $\chi^2_{8,0.025} = 2.180$ and $\chi^2_{8,0.975} = 17.535$ to derive and calculate a central 95% confidence interval for σ^2. **[4 marks]**

Student response

(a)

(b)

(c)